

Autosegmental Analysis of Morphologically Conditioned Epenthetical Processes in Tiv NounsFrancis Iorchia Ayagah¹

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Email: tagesa@gmail.com**ABSTRACT**

This study examines the morphologically conditioned epenthetical processes in Tiv nouns. The study seeks to find out which environments epenthetic processes occur in Tiv nouns, what rules or constraints govern the processes and the implications of these on Tiv grammar. To achieve these, the study adopts the qualitative research design. Data was collected from monolingual native speakers of the language through participant observation, non-directive interviews and relevant scholarly materials. Data collection was restricted to Gboko, whose speech norms are considered the standard variety. The study employed Leipzig glossing rules for organising and coding data. The deductive method of data analysis was adopted since the study is theory-guided. Data was analysed using an auto-segmental theoretical approach, and findings show that epenthetic processes are morphologically conditioned in that they occur through affixation in pluralisation. This study discovers that epenthesis in Tiv occurs in Nouns at the initial, medial and final positions. It reveals that the common epenthesis in nouns at initial positions in Tiv are vowels (henceforth vowel epenthesis), glide epenthesis occupies medial position in Tiv while consonants take the nouns' final position. The study further portrays that the processes of epenthesis in Tiv are governed by the Tiv phonotactics. The study concludes that inappropriate application of epenthetic sound could lead to poor reception and, consequently, no comprehension. The study calls for more research in morphophonemics, considering the fact that the outputs of morphological processes are inputs for phonological outcomes and thus an unavoidable need for an interface between planes of linguistic analysis.

Key words: Tiv, Nouns, Epenthesis, morphologically conditioned, auto-segmental and phonotactics

Introduction

Morphophonemic studies reveal that the outputs of morphological processes constitute the inputs for phonological outputs. The morphologically conditioned phonological processes are those phonological processes which result from morpheme concatenation. When morphemes are concatenated, new phonological environments are created resulting in numerous phonological processes. This study examines epenthesis as one of the morphologically conditioned phonological processes in Tiv with particular emphasis on Tiv nouns.

Epenthesis in Tiv nouns is a morphologically conditioned phonological process in that it is a consequence of affixation. The output of affixation in Tiv nouns conditions phonological outputs whereby one or more sounds do not present originally are introduced either at the beginning, middle or end of a noun as specified by the phonotactic requirements of the language. This needs proper scholarship because improper pronunciation of words affects reception by listeners, with a consequential impediment of comprehension and interpretation.

What informs this study is the fact that the existing studies on Tiv have revealed that less attention is paid to sound changes which are occasioned by morphologically conditioned phonological processes in the language generally, and epenthesis, and this leads to a dearth of literature in the area. It is against this background that this study seeks to investigate epenthesis as a morphologically conditioned phonological process to reveal its patterns and describe the environments.

Review of Related Literature

Meaning of Epenthesis

Phonological scholars have considered epenthesis from different perspectives. Some see it as a purely phonological process, while others consider it a consequence of the interface between morphology and phonology. Crystal (2008, p.171), for instance, defines epenthesis as a “term used in phonetics and phonology to refer to a type of intrusion, where an extra sound has been inserted in a word. The definition considers epenthesis as an exclusive preserve of phonetics and phonology without taking into cognisance the morphological situations that condition epenthetic sounds.

Hall (2011) looks at epenthesis as a process where a vowel is added to a word to repair an input that does not match the structure of a language, or to allow the syllabification of stray consonants. This study finds the definition of epenthesis as given by Hall (2011) to be too simplistic. The present study sees epenthesis beyond the insertion of vowels to a word, considering the fact that in Tiv, both vowels and consonants are found in epenthetic patterns.

To Jaradat (2019), epenthesis is the process of inserting a vowel or a consonant to avoid creating syllables which are not part of the syllable templates in a certain language or dialect. The definition by Jaradat (2019) is acceptable to some extent. However, the question posed by this

study, which is not addressed in the definition, is whether the insertion is purely phonological, morphological or a product of an interface between morphology and phonology.

In the context of this study, therefore, epenthesis is considered as a product of the interface between morphology and phonology. The analysis of epenthesis may vary depending on the theory applied or the perspective of the study. In this study, epenthesis is considered a phonological process which is conditioned by morpheme concatenation

Classification and functions of epenthesis

The classification of epenthesis, just like its definition, is done differently by different scholars. For instance, Crystal (2008, p.171) classifies epenthesis into two subclasses, namely prothesis (the insertion of sound/s at the beginning of a word) and anaptyxis (the insertion of a sound in between consonants). This classification is not comprehensive enough. The epenthetic process varies from one language to another, with varying functions.

Aor (2020) classifies epenthesis in four main types as follows:

Prosthesis: - the insertion of a sound at the beginning of words without changing the word, as in
 ‘pyaven’, ‘ => upyaven,
 ‘sentimita’ => usentimita’,
 ‘erdoo’ => uerdoo.

Anaptyxis (the addition of an extra vowel between two consonants) as shown in the words:

Kirimishi => [Christmas],
 igipiti => [Egypt],
 bokoti => [bucket],
 tabul => table]

Excrecence (addition of a consonant in between other consonants to ease pronunciation) as in:

ihwa => ihywa,
 kwande => kywande,
 iwen => iywen,
 iwanger => iywanger

Paragoge (the addition of a sound (mostly a vowel) at the end of a word). According to Aor, a typical example of a consonantal paragoge is the addition of [h], while the commonest paragogic vowels in Tiv language are *u*, *i*, *o*, and *a*.

The classification of epenthesis by Aor reflects all the positions in which epenthesis occurs cross-linguistically. However, his examples do not reconcile with his classification in that there is a radical change in the morphological makeup of the borrowed words and their Tiv version beyond mere insertion of letters/sounds. For instance, the English word *Chrsitmas* and its Tiv version *Kirimishi* is rather a case of transmorphemisation - a kind of substitution in which certain morphemes from the donor language are substituted to condition adaptation in consonance with the phonotactics of the borrowing language (Filipović, 1980).

This study argues that the phonotactics of Tiv truly do not permit consonant clusters. Consequently, any borrowed word into the language with a consonant cluster must be made to

conform to its phonotactic requirements. This study, therefore, submits that the morphophonological process through which words borrowed from English are repaired to conform to the phonotactic requirement of the Tiv language is transmorphemisation and not epenthesis, as Aor (2020) misconstrues. However, the sub-classification of epenthesis by Aor gives this present an insight into the epenthetic environments in Tiv.

Žygis (2000) classifies insertions (epenthesis) into three main types: grammatical, phonetic and prosodic. He further explicated that grammatical insertions comprise all types of insertions which are conditioned morphologically and syntactically. Žygis asserts that several insertions already take place in the grammar of a given language. They are idiosyncratically characteristic of a particular language. Insertions of this type are subject to phonological, morphological or syntactic regularities and are determined strictly by grammatical categories or constituents. He contends further that morphological insertions take place within words, and they are mainly between prefixes and stems or stems and suffixes and between constituents of compounds, while syntactic insertions occur between words, that is, at a syntactic level.

A good instance in Žygis (2010) is the [t]-insertion in Axinica Campa, which is morphologically restricted: it takes place only in suffixation processes, as shown in the following examples:

Axinica Campa			
/i-N-koma-i/	→	[iŋkomati]	‘he will paddle’
/i-N-koma-aa-i/	→	[iŋkomataati]	‘he will paddle again’ iwanger iywen

Žygis (2010) further asserts that frequent epenthetical processes include glide insertions, which, on the one hand, take place in a well-defined morphological context and, on the other hand, might be governed phonologically. For example, in Sinhala, the underlying vowel hiatus at root-suffix boundaries is always resolved, whereby the type of repaired strategy depends on the lexical category of the root. For instance, the hiatus in nouns is always resolved by glide insertion, as shown in the following examples:

Sinhala			
/ræ+a/	[ræjə]	‘night, sg.def.’	
/toppi+a/	[toppijə]	‘hat, sg.def.’	
/ašu+a/	[ašuwə]	‘attic, sg.def.’	
/maaligaa+a/	[maaligaawə]	‘palace, sg.def.’	

(Smith 2001:63 in Žygis, 2010)

Phonetic insertions considerably differ from grammatical ones as they emerge on the surface representation as a natural consequence of the interaction of phonetic factors. Therefore, in contrast to grammatical insertions, they are solely explicable in phonetic terms. This type of insertion comprises a wide spectrum of possible sounds; the occurrence of which depends exclusively on the neighbouring sounds (Kawahara, 2002, cited in Žygis, 2010, p.120).

Phonetic insertions are often gradual processes evolving from overlapping gestures, which at the beginning may not be perceived as a categorical sound.

[w] and [j]-insertions in Japanese

guai gu[w]ai ‘condition’

siawase si[j]awase ‘happiness’

(Kawahara 2002 cited in Žygis 2010, p.120)

Prosodic insertion is primarily prosodically conditioned as its appearance is determined by prosodic constituents such as syllable, prosodic foot, prosodic word, phonological phrase, intonational phrase, and phonological utterance. The insertions are found either at the boundaries of the prosodic constituents or they require a given prosodic constituent as a domain of their application. Since prosodic constituents are often domains of stress assignment, prosodic insertions can also interact with stress

This study aligns itself with Žygis (2010) in terms of functional classification of epenthesis. However, it can be faulted as the analysis is seemingly limited to grammatical insertions since the thesis of Žygis’ classification is the analysis of morphologically conditioned epenthetical processes in Tiv nouns.

To Samuels (2017: p. 93), epenthesis arises for a variety of reasons. The phonotactic requirements of a given language may discourage vowels in hiatus or consonant clusters, and a consonant or vowel may be added to make pronunciation easier. The author outlines the following functions of epenthesis: (a). separating vowel: A consonant may be used or added to separate vowels in a hiatus. This is the cause with [Linking and intrusive R] ----- "drawing"-----"draw-r-ing". (b). breaking consonant clusters: A vowel may be placed between consonants to separate them. Hamtramck ----- Hamtramick. Regular examples in English are /i/ used in forming Latinate words such as equidistant, and /o/, used in forming words on Greek roots or general compounds, as in speedometer. (c). bridging consonant clusters: A consonant may be placed between consonants in a consonant cluster where the place of articulation is different. For example, where one consonant is labial and the other is alveolar. -Something ----something -a-mrotos----umbrotos.

The functions of epenthesis enumerated by Samuels (2017: 93) are acceptable, but this study has noticed that they are all within the realms of phonologically conditioned epenthesis in that the functions are phonetic and prosodic rather than grammatical. These are in contrast with grammatical epenthesis, which this study interrogates and therefore gives room for more investigation.

Morley (2022), in her study entitled *Consonant epenthesis* argues that epenthesis is used to refer to the addition of segmental material to satisfy a morphological template or minimal word length requirement. According to the author, theoretically, epenthesis may occur as the result of a phonological, morphological, or phonetic rule. Within generative theory, epenthesis is “triggered” or “conditioned” by the presence of specific environments. Such environments may consist of sequences that are disallowed or dis-preferred within the language (*XY), and that are prevented from surfacing by the operation of epenthesis (/XY/) [XBY]).

This study agrees with Morley (2022) that epenthesis is meant to satisfy a morphological template or minimal word length. This study opines that morphology is the engine room for

phonological processes. The satisfaction of morphological template conditions phonological outputs which satisfy phonotactic constraints in any given language

Methodology of the study

This study adopts the qualitative research design. Data was collected from monolingual native speakers of the language through participant observation, non-directive interviews and relevant scholarly materials. Data collection was restricted to Gboko, whose speech norm is considered the standard variety. The study employed Leipzig glossing rules for organising and coding data. Deductive method of data analysis was adopted since the study is theory-guided. Data was analysed using an autosegmental theoretical approach

Theoretical framework

Autosegmental Phonology is a framework of phonological analysis proposed by John Goldsmith in his PhD thesis in 1976. The central tenet of auto-segmental phonology is that sounds or with a technical term, segments (or phonemes) can be split up into smaller units, which can be independently manipulated by phonological operations such as deletion, tone, insertion, and spreading. Goldsmith (1990: p. 8) explains the different approach of this theory from the generative method of representation.

Autosegmental representation differs from familiar generative and traditional phonemic representation in that it consists of two or more tiers of segments. In the picture given us by classical generative phonology - and, indeed, most theories of phonology and phonological representation – phonological representations consist of a string of segments. In autosegmental representation, however, we posit two or more parallel tiers of phonological segments. Each tier itself consists of a string of segments, but the segments on each tier differ with regard to what features are specified in them.

What this study extrapolates from Goldsmith (1990: 8) is that within Autosegmental phonology, each tier consists of segments that are independent of the other segments in another tier. Each tier is associated with the other level of tier through the means of association lines, which are governed under the Association Convention. In this study, this convention is modified to affect the principle of well-formedness. The study employs three types of tiers due to the modification, namely, the epenthetic tier, the CV tier, and the Segmental tier. Tonal tier is, however, excluded from the analysis of the data.

Data presentation and analysis

Prosthesis in Tiv

A prosthesis is the addition of one or more sounds

The data collected from the field indicate that there are five morphemes which condition prosthesis in Tiv. These include ['i-', 'a-', 'u-', 'm-' and 'mba-']. While ['i-', 'a-', 'u-' 'mba-'] are bound morphemes which are fused to lexical stems or base morphemes to transform the singular

state of nouns, into plural forms. ['m-'] is added to verb stem to derive nouns as shown in the data below:

Verb	Gloss	Noun	gloss
Hen	think	mhen	thought
Yem	go	myem (gerund)	the going
(a)	àngèv illness	í.àngèv illnesses	
insertion tier	$\left(\begin{array}{ccccc} -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} \\ & & & & \\ \text{v} & \text{c} & \text{c} & \text{v} & \text{c} \end{array} \right) \rightarrow \left(\begin{array}{cccccc} +\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} \\ & & & & & \\ \text{v} & \text{v} & \text{c} & \text{c} & \text{v} & \text{c} \end{array} \right) \rightarrow [\text{V.}]$		
CV tier	$\left(\begin{array}{ccccc} & & & & \\ & & & & \\ \text{a} & \text{n} & \text{g} & \text{e} & \text{v} \end{array} \right) \rightarrow \left(\begin{array}{cccccc} & & & & & \\ & & & & & \\ \text{i} & \text{a} & \text{n} & \text{g} & \text{a} & \text{v} \end{array} \right)$		
Segmental tier			

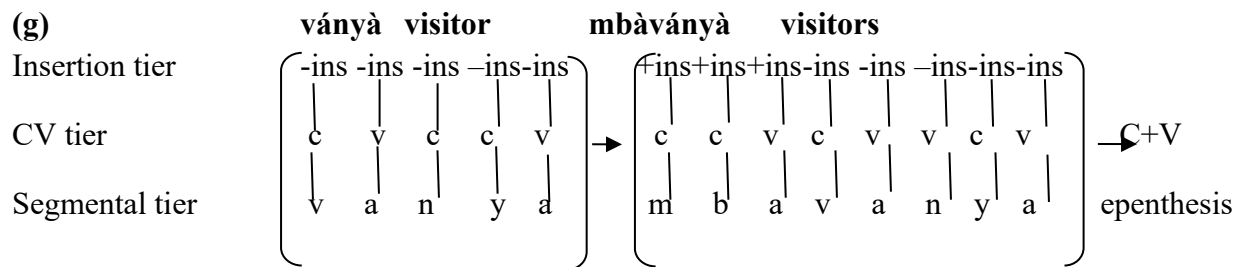
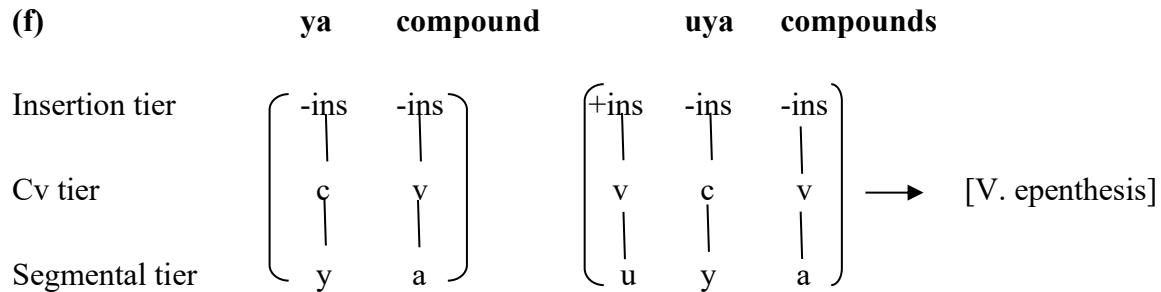
(b)	ōr [person]	í-ōr [persons]
Insertion tier	$\left(\begin{array}{cc} -\text{ins} & -\text{ins} \\ & \\ \text{v} & \text{c} \end{array} \right) \rightarrow \left(\begin{array}{ccc} +\text{ins} & -\text{ins} & -\text{ins} \\ & & \\ \text{v} & \text{v} & \text{c} \end{array} \right) \rightarrow [\text{V. epenthesis}]$	
CV tier	$\left(\begin{array}{cc} & \\ & \\ \text{o} & \text{r} \end{array} \right) \rightarrow \left(\begin{array}{ccc} & & \\ & & \\ \text{i} & \text{o} & \text{r} \end{array} \right)$	
Segmental tier		

(c)	hir [talisman]	a-hir [talisman]
insertion tier	$\left(\begin{array}{ccc} -\text{ins} & -\text{ins} & -\text{ins} \\ & & \\ \text{c} & \text{v} & \text{c} \end{array} \right) \rightarrow \left(\begin{array}{cccc} +\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} \\ & & & \\ \text{v} & \text{c} & \text{v} & \text{c} \end{array} \right) \rightarrow [\text{V. epenthesis}]$	
CV tier	$\left(\begin{array}{ccc} & & \\ & & \\ \text{h} & \text{i} & \text{r} \end{array} \right) \rightarrow \left(\begin{array}{cccc} & & & \\ & & & \\ \text{a} & \text{h} & \text{i} & \text{r} \end{array} \right)$	
Segmental tier		

(d)	bamber [piece]	a-bamber- [pieces]
insertion tier	$\left(\begin{array}{ccccccc} -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} \\ & & & & & & \\ \text{c} & \text{v} & \text{c} & \text{c} & \text{v} & \text{c} & \text{c} \end{array} \right) \rightarrow \left(\begin{array}{cccccccc} +\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} \\ & & & & & & & \\ \text{v} & \text{c} & \text{v} & \text{c} & \text{c} & \text{v} & \text{c} & \text{c} \end{array} \right) \rightarrow [\text{V.}]$	
CV tier	$\left(\begin{array}{ccccccc} & & & & & & \\ & & & & & & \\ \text{b} & \text{a} & \text{m} & \text{b} & \text{e} & \text{r} & \end{array} \right) \rightarrow \left(\begin{array}{cccccccc} & & & & & & & \\ & & & & & & & \\ \text{a} & \text{b} & \text{a} & \text{m} & \text{b} & \text{e} & \text{r} & \end{array} \right)$	
Segmental tier		

(e).	zwar [reservoir]	u-zwar [reservoirs]
insertion tier	$\left(\begin{array}{cccc} -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} \\ & & & \\ \text{c} & \text{c} & \text{v} & \text{c} \end{array} \right) \rightarrow \left(\begin{array}{ccccc} +\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} & -\text{ins} \\ & & & & \\ \text{v} & \text{c} & \text{c} & \text{v} & \text{c} \end{array} \right) \rightarrow [\text{V. epenthesis}]$	
CV tier	$\left(\begin{array}{cccc} & & & \\ & & & \\ \text{z} & \text{w} & \text{a} & \text{r} \end{array} \right) \rightarrow \left(\begin{array}{ccccc} & & & & \\ & & & & \\ \text{u} & \text{z} & \text{w} & \text{a} & \text{r} \end{array} \right)$	
Segmental tier		

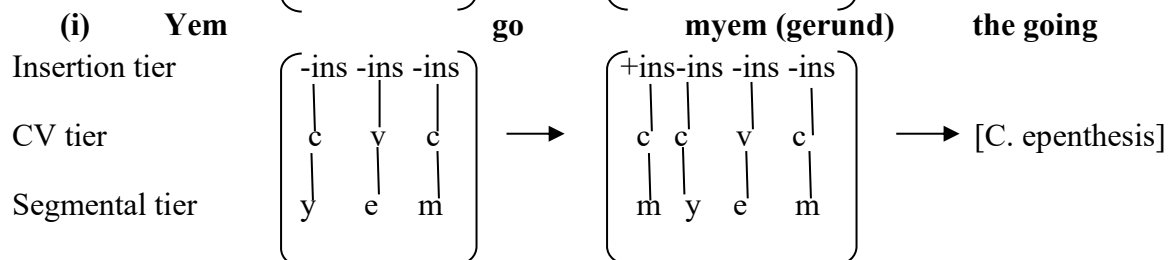
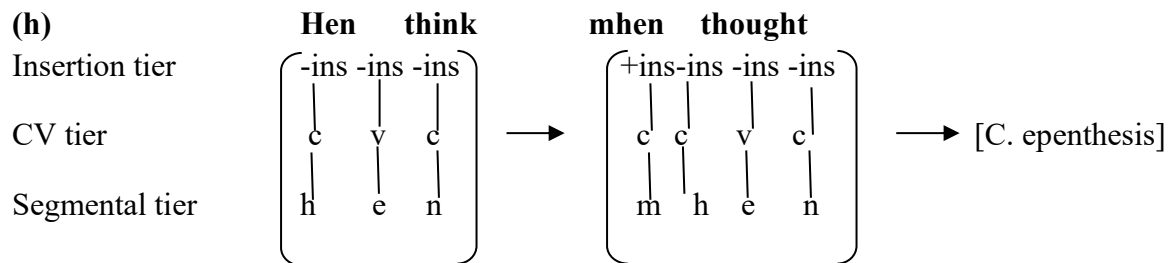
Segmental tier z w a r u z w a r



In (g) above, two sounds are added. The [m] and [b] are realized as a single sound in Tiv. Ayagah and Ugechi (2021, p.61) explain thus:

Tiv onset has a moderately complex syllable structure without a consonant cluster. The onset constituent of the Tiv syllable is made up of a single consonant segment or complex segments, which are counted phonologically as single segments, but which have internal structures comparable to sequences of segments (clusters).

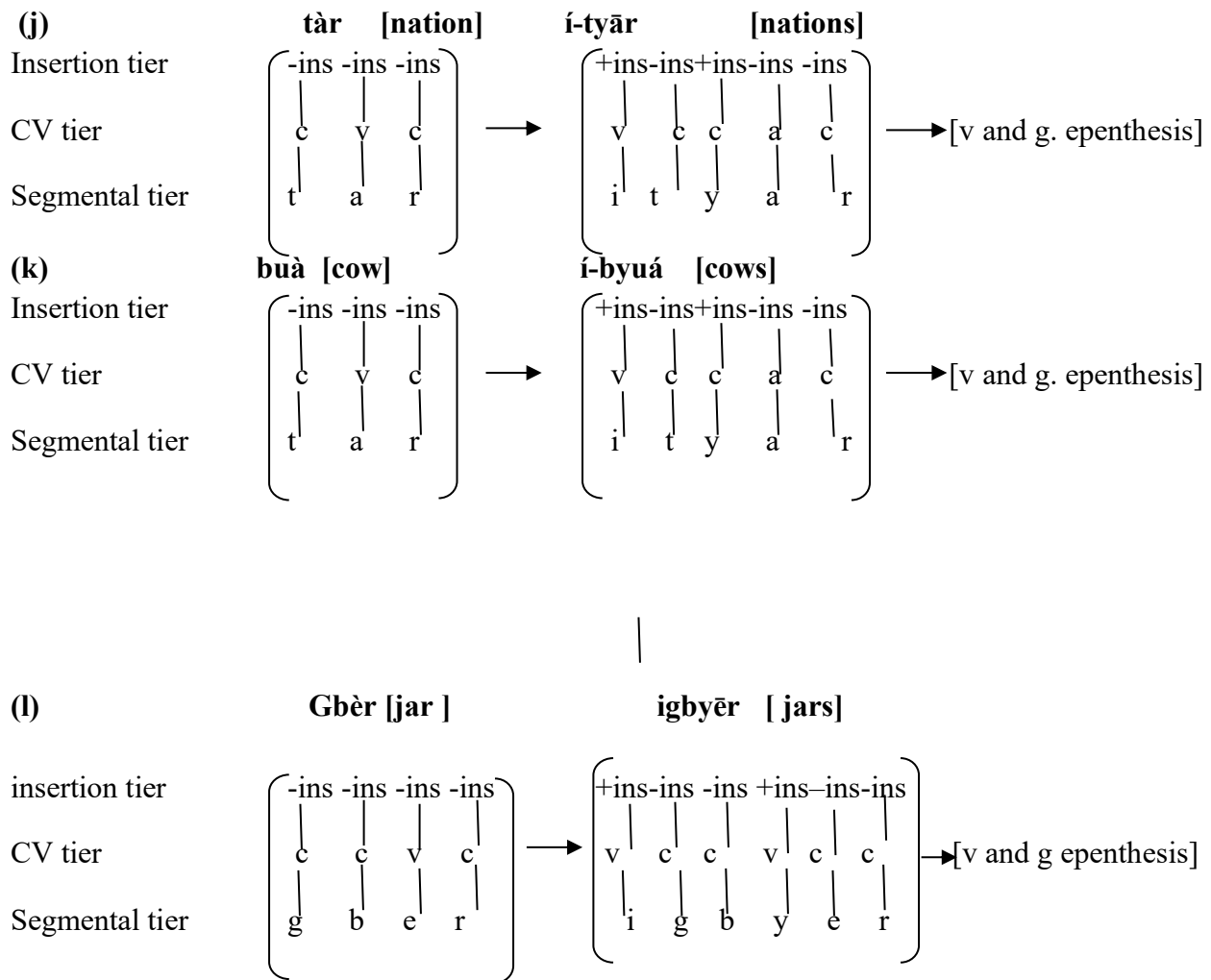
Based on the above explanation, this study asserts that the epenthetic pattern in (g) is a prosthesis with two sounds, the consonant [mb] and the vowel [a]. The [mb] internal structures show a sequence of segments that is counted phonologically as a single segment, considering the fact that Tiv has co-articulations but not a consonant cluster.



Unlike the prosthesis in a-g which are added to a Tiv noun to inflect it for plural, the bilabial nasal sound [m] is added to a Tiv noun for derivational purposes (to change a verb to noun) as shown in (h) and (i).

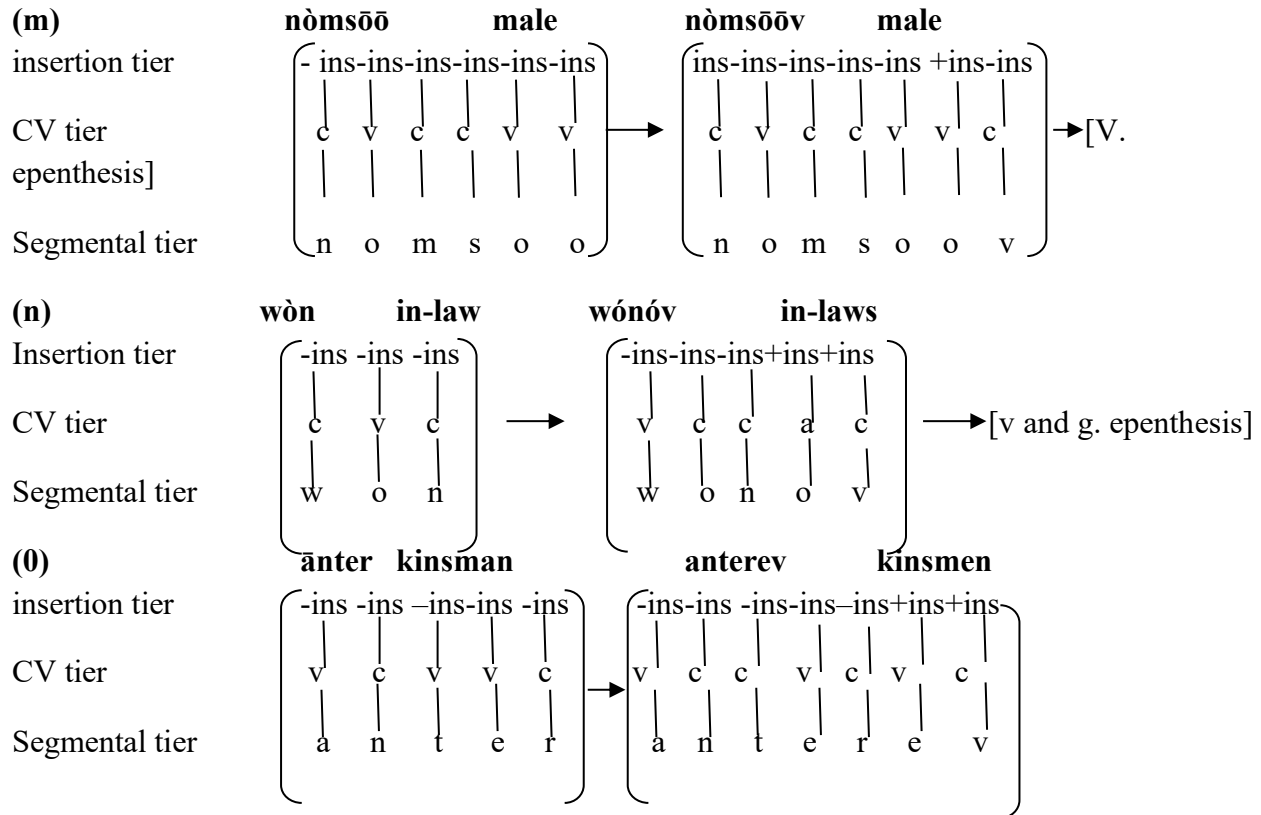
Glide epenthesis

Morphologically induced glide epenthesis in Tiv noun occurs in environments where the front high unrounded vowel /i/ is inserted as prosthesis to the nominal base whose initial consonants is any of the following /p/, /b/, /t/, /d/, /k/, /kp/, /g/, /gb/, /f/, /v/, /m/, /mb/, /ɲ/, /h/ and /l/. Glide insertion in Tiv nouns is for grammatical purposes. The insertion of [i] at the initial position and [j] at the medial position of a Tiv noun changes it from singular to plural. Below is auto-segmental analysis of glide.



Paragoge

It is a type of epenthesis in which one or sounds are added at the end of a word. Paragoge in Tiv nouns is restricted to the following sounds ‘v’ ‘ev’ ‘ôv’ ‘ov’ and ‘m’ as analysed below



Summary of Findings

This study finds that epenthesis in the Tiv nouns is a morphologically conditioned phonological process in that is conditioned by the process of morphology of affixation. The study reveals that epenthetic patterns in Tiv nouns inflect the nouns for plurality. Also discovered in the study is the fact that there are five morphemes ['i-', 'a-', 'u-', 'm-' and 'mba-'] which condition prosthesis in Tiv. That out of these five morphemes, ['i-', 'a-', 'u-' 'mba-'] are bound morphemes which are fused to lexical stems or base morphemes to transform the singular state of nouns, into plural forms while ['m-'] is added to verb stems to derive nouns. When, however, it functions as a prosthesis and a paragoge, it drops its derivational functions to assume the inflectional function of transforming a noun from singular form to plural form. The study also discovers that anatyxis and excrescence are not common in Tiv nouns except in borrowed nouns. The study however insists that Tiv nouns which occur in environments where the front high unrounded vowel /i/ is inserted as prosthesis to the nominal base whose initial onset is any of the following /p/, /b/, /t/, /d/, /k/, /kp/, /g/, /gb/, /f/, /v/, /m/, /mb/, /ŋ/, /h/ condition the addition of a medial glide between the onset of the second syllable and its nucleus. The analysis reveals that paragoge in Tiv nouns is restricted to the following sounds: 'v', 'ev', 'ôv', 'ov', and 'm.' It could, therefore, be rightly concluded here that the inappropriate application of epenthetic sound could lead to poor reception and consequently impede effective comprehension.

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